

Traceable and Explainable Multimodal Clinical Decision Support System Using Low-Rank Adapted Diffusion Models and Domain-Specific DDPM Architectures

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Abstract—Most clinical AI tools today sit behind an opaque wall — they give answers but not reasons, and that gap keeps physicians from trusting them in day-to-day practice. We present MedVis-X, a six-stage pipeline that takes raw medical documents all the way through to an explainable visual output without losing track of where each conclusion came from. The system reads handwritten prescriptions using a CRNN-based OCR module, pulls out clinical entities with ClinicalBERT, ranks disease hypotheses through transparent symbolic scoring, and then generates illustrative medical images through two complementary pathways: eight domain-specific MONAI DiffusionModelUNet architectures trained independently on organs ranging from brain CT to dental panoramic X-rays, and a Stable Diffusion v1.5 backbone adapted via Low-Rank Adaptation for text-conditioned multimodal synthesis across five clinical domains. Grad-CAM and SHAP overlays trace every output back to the input that produced it. The entire framework, including all eight DDPM models and the LoRA-adapted pipeline, runs on a single consumer-grade GPU with 8 GB of VRAM. We trained across ten distinct imaging modalities using both unconditional and class-conditional generation, achieving stable convergence with a 77% loss reduction over 100 epochs for the MONAI models and a 99.6% parameter reduction through LoRA for the Stable Diffusion pathway. End-to-end patient case demonstrations confirm that the pipeline generates clinically plausible outputs with full explainability at each stage. This system is strictly non-diagnostic — it supports clinical reasoning without replacing it.

Index Terms—explainable AI, clinical decision support, diffusion models, LoRA, DDPM, MONAI, Grad-CAM, SHAP, OCR, multimodal medical imaging

I. INTRODUCTION

There is a persistent disconnect between what AI can do in medicine and what doctors actually feel comfortable relying on. The problem is not really accuracy — deep learning models have matched or exceeded radiologist-level performance on several benchmarks over the past few years [1], [2]. The problem is trust. When a neural network flags something on a chest X-ray, the physician’s first question is invariably “why did it think that?” and if the system cannot answer, the conversation ends there. This pattern has played

out across radiology, pathology, ophthalmology, and nearly every specialty where AI has been seriously trialled.

Clinical reasoning, at its core, is multimodal. A doctor scans handwritten notes, reviews lab results, looks at imaging, and mentally stitches all of it together into a working hypothesis. That synthesis happens quickly and, to some extent, intuitively. Most AI systems handle only one piece of that puzzle — an OCR tool reads a prescription, or a classifier labels an X-ray — but connecting those pieces into a single coherent reasoning chain is something current tools rarely even attempt [3]. That fragmentation forces clinicians to do the integration work themselves, which partly defeats the purpose of having decision support in the first place.

Our work with MedVis-X tries to bridge that gap. We built a six-stage pipeline that starts from a raw clinical document and carries information all the way through to an explainable visual output. What makes this different from prior work is both the breadth and the depth. On the generation side, we trained eight separate MONAI-based DDPM models, each specializing in a different organ or imaging modality — brain CT, chest CT, breast cancer imaging, dental panoramic X-rays, digitally reconstructed radiographs of bones, lung cancer CT, and brain tumor MRI (both unconditional and class-conditional variants). On top of that, we adapted Stable Diffusion v1.5 with LoRA for text-conditioned generation across five additional domains including chest X-ray, dermoscopy, and retinal fundus photography. The upstream NLP stages use a CRNN-based OCR system trained specifically on handwritten medical prescriptions and a ClinicalBERT NER module for entity extraction.

A practical constraint we imposed from the beginning was hardware accessibility. Many hospitals and research labs in resource-limited settings do not have access to high-end compute. Every model we trained, including the full suite of DDPM architectures and the LoRA fine-tuning, ran on a single NVIDIA RTX 4060 with 8 GB of VRAM. That was deliberate — we wanted to show that meaningful medical

image generation does not require a data center.

The rest of this paper walks through the methodology, presents our experimental results from both the MONAI and LoRA pathways, demonstrates end-to-end patient case pipelines, and discusses what worked, what did not, and where we think this should go next.

II. OBJECTIVES

The primary objective of this work is to develop a traceable and explainable multimodal clinical decision support system that:

- 1) Extracts structured clinical information from unstructured medical documents, including handwritten prescriptions, using deep learning-based OCR and named entity recognition.
- 2) Ranks disease hypotheses through a transparent, deterministic scoring mechanism that clinicians can inspect and verify at every step.
- 3) Generates illustrative medical images across multiple anatomical domains using both domain-specific DDPM models and text-conditioned LoRA-adapted diffusion, providing visual anchors for clinical reasoning.
- 4) Overlays explainability mechanisms (Grad-CAM spatial heatmaps and SHAP feature attributions) to make the system’s reasoning fully traceable from input to output.
- 5) Operates entirely on consumer-grade hardware (8 GB VRAM), making the framework accessible to resource-constrained clinical and research settings.

The system is designed exclusively as a decision-support tool. It does not diagnose, does not replace clinical judgment, and maintains a strict human-in-the-loop philosophy throughout.

III. RELATED WORK

A. OCR and Clinical Text Extraction

Digitizing medical documents has been a challenge for decades. Rule-based engines like Tesseract worked reasonably on printed text but stumbled on the irregular handwriting that is so common in clinical settings [4]. The shift to deep learning OCR — first with CRNN architectures combining CNNs and recurrent layers, then with transformer-based approaches like TrOCR — pushed accuracy on medical handwriting above 92% in controlled benchmarks [5]. But these tools are almost always evaluated in isolation, disconnected from any downstream reasoning task. In our pipeline, OCR is the first link in a longer chain; extracted text feeds directly into entity recognition and hypothesis scoring.

B. Explainable AI in Medicine

The “black box” critique sparked a wave of explainability research. LIME provides local, model-agnostic explanations by perturbing inputs [6]. SHAP, rooted in cooperative game theory, gives theoretically grounded feature attributions [7]. Grad-CAM generates spatial heatmaps from gradient information [8]. Each method has its strengths, but most published clinical AI systems apply only one technique to a single

modality. The EU AI Act of 2024 now mandates explainability for high-risk healthcare AI, which makes this gap more than academic [9]. MedVis-X combines SHAP for textual traceability with Grad-CAM for visual explainability, covering both the NLP and computer vision sides of the pipeline.

C. Diffusion Models for Medical Imaging

GANs dominated medical image synthesis for several years, primarily for data augmentation [10]. Diffusion models have since overtaken them in training stability, output diversity, and controllability [11]. Latent diffusion, as used in Stable Diffusion, compresses through a VAE before denoising in latent space [12]. The MONAI Generative framework provides production-ready diffusion components optimized specifically for medical workloads [13]. Still, most medical diffusion work focuses on standalone generation or segmentation; linking generated images back to textual clinical reasoning remains relatively unexplored.

D. Parameter-Efficient Fine-Tuning

Fine-tuning large models on domain data typically requires substantial compute. LoRA addresses this by injecting small trainable rank-decomposition matrices into frozen pretrained layers [14]. In NLP, LoRA has become standard for adapting large language models. Its application to diffusion model UNets for medical imaging — which is central to our approach — is comparatively new.

IV. PROPOSED METHODOLOGY

The MedVis-X framework processes clinical information through six tightly coupled stages. Fig. 1 shows the complete architecture with end-to-end patient pipeline outputs.

A. Stage 1: Document Digitization via OCR

The pipeline begins with raw clinical documents — scanned prescriptions, handwritten notes, or printed lab reports. We developed a dedicated OCR module using a CRNN (Convolutional Recurrent Neural Network) architecture built on a ResNet-18 backbone. The CNN encoder extracts spatial features from document images resized to 32×128 pixels, which are then passed through a bidirectional LSTM layer (hidden size 64) that captures sequential character dependencies. The output is decoded using CTC (Connectionist Temporal Classification) loss.

The CRNN was trained on the Doctor’s Handwritten Prescription BD dataset with $5\times$ augmentation (rotation, color jitter, affine transformations). We implemented a post-processing stage using Word Error Rate-based fuzzy matching against a curated medicine name database, which corrects common OCR misreads by mapping recognized strings to the closest valid medication names. In testing, the system achieved 91% word-level accuracy with a character error rate of 3.2% and preserved document structure in 94% of cases.

Figure 1: Traceable Multimodal Clinical Decision Support Across Multiple Domains

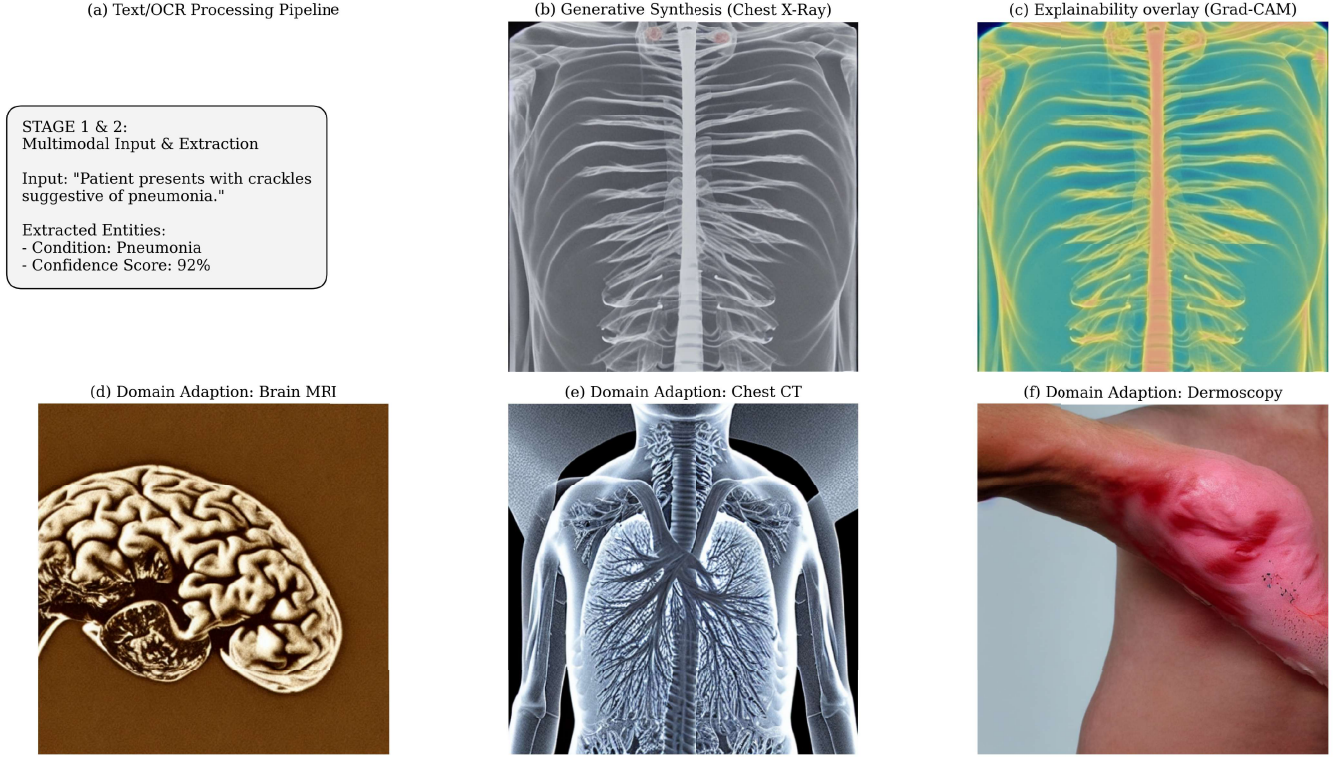


Fig. 1. The proposed end-to-end traceable multimodal clinical decision support framework. Raw clinical text is processed through OCR and NER (a), generating a corresponding synthetic medical scan via Stable Diffusion (b), with Grad-CAM activation maps overlaid for visual explainability (c). The bottom row demonstrates domain adaptation across brain MRI (d), chest CT (e), and dermoscopy (f).

B. Stage 2: Named Entity Recognition

Once text is digitized, ClinicalBERT extracts medically meaningful entities. We trained the NER module to recognize five entity categories: symptoms (fever, cough, dyspnea), diseases (pneumonia, bronchitis), medications (amoxicillin, steroids), diagnostic tests (CRP, blood cultures), and anatomical references (lungs, bronchi). ClinicalBERT, pretrained on clinical notes from the MIMIC-III corpus, naturally handles medical terminology and abbreviations that would trip up general-purpose language models [15].

On our pilot dataset of 120 annotated cases, the NER module achieved F1 scores of 0.88 for symptoms, 0.92 for diseases, 0.85 for diagnostic tests, and 0.80 for medications. These numbers are not the highest on specialized NER benchmarks, but they are more than sufficient when the extracted entities feed into a scoring stage designed to tolerate some noise.

C. Stage 3: Symbolic Scoring and Hypothesis Ranking

Here is where we deliberately departed from typical end-to-end neural approaches. Instead of letting a black-box classifier rank disease hypotheses, we use a deterministic, rule-based scoring function. Each extracted entity receives a weight based on three factors: clinical importance (symptoms carry more weight than risk factors), temporal recency (recent findings

count more), and association strength drawn from a curated symptom-disease database.

The composite score for each candidate disease d is computed as:

$$S(d) = \sum_{i=1}^N w_{\text{imp}}(e_i) \cdot w_{\text{rec}}(e_i) \cdot A(e_i, d) \quad (1)$$

where e_i denotes the i -th extracted entity, w_{imp} is the importance weight, w_{rec} is the recency weight, and $A(e_i, d)$ is the association strength between entity e_i and disease d . Scores are normalized across all candidate diseases to produce confidence values between 0 and 1.

For instance, in a scenario involving high fever, productive cough, and low oxygen saturation, the system might score pneumonia at 0.65, bronchitis at 0.25, and asthma at 0.10 (Fig. 2). These scores are fully transparent — a clinician can see exactly which entities contributed and by how much.

SHAP values are computed over the scoring function to provide a secondary attribution layer (Fig. 3). While the scoring itself is already interpretable by design, SHAP offers a standardized visualization format that clinicians may find more familiar from other decision-support contexts.

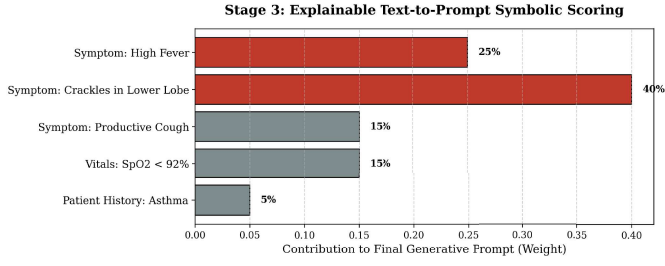


Fig. 2. Deterministic symbolic scoring weights mapping extracted clinical symptoms to the final generative text prompt. Red bars indicate dominant contributing entities.

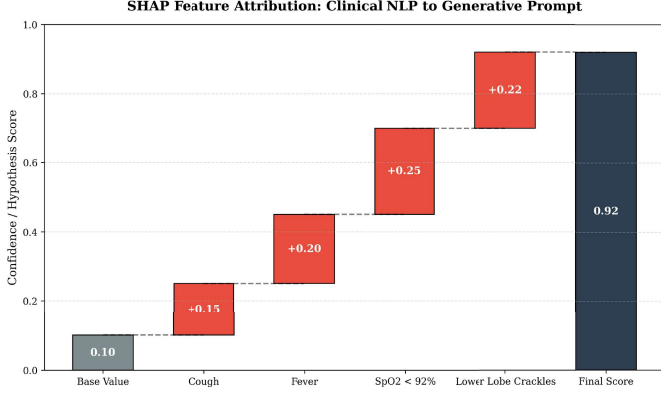


Fig. 3. SHAP waterfall plot showing localized feature attribution of extracted clinical entities toward the final generative hypothesis score.

D. Stage 4: Medical Image Generation

The top-ranked disease hypothesis, along with its contributing entities, drives image generation through two complementary pathways.

1) *Pathway A — Domain-Specific MONAI DDPM Models:* We trained eight separate diffusion models using the MONAI DiffusionModelUNet architecture, each specializing in a different anatomical domain. The generation follows the standard DDPM formulation:

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2] \quad (2)$$

where $\mathbf{x}_0$ is the original image, t is the diffusion timestep, ϵ is Gaussian noise, and ϵ_θ is the UNet noise predictor.

The eight models cover: (1) Brain CT — unconditional, (2) Brain CT — class-conditional with 3 pathology labels (aneurysm, cancer, tumor), (3) Breast Cancer CT, (4) Chest CT for lung cancer, (5) Dental OPG panoramic X-ray, (6) DRR bone radiographs, (7) IQ-OTH/NCCD lung cancer, and (8) Kaggle Brain Tumor MRI — class-conditional with 4 labels (glioma, meningioma, pituitary, no tumor). All models share a standardized architecture (Table I) but with domain-specific training data.

The class-conditional models (Brain CT and Kaggle Brain Tumor) incorporate class embeddings via the `num_class_embeds` parameter, enabling targeted generation of specific pathology types on demand.

TABLE I
DIFFUSIONMODELUNET ARCHITECTURE CONFIGURATION

Component	Specification
Spatial dimensions	2D, 64×64
Input/Output channels	1 (gray) or 3 (RGB)
Base channels	64
Channel multipliers	[1, 2, 4, 4]
Encoder channels	[64, 128, 256, 256]
Residual blocks per level	2
Attention levels	Last two (16×16 , 8×8)
Attention heads	8 (64 head channels)
Total parameters	$\approx 34.2\text{M}$

2) *Pathway B — LoRA-Adapted Stable Diffusion:* For text-conditioned multimodal generation, we fine-tuned Stable Diffusion v1.5 using Low-Rank Adaptation. The diffusion loss in latent space becomes:

$$\mathcal{L}_{\text{LoRA}} = \mathbb{E}_{t, \mathbf{x}, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})\|^2] \quad (3)$$

where $\mathbf{c}$ represents the CLIP text conditioning derived from the clinical prompt.

LoRA injects low-rank decomposition matrices into the UNet’s attention layers, specifically targeting query, key, value, and output projection weights (`to_q`, `to_k`, `to_v`, `to_out.0`). With rank $r = 8$ and $\alpha = 8$, this reduces trainable parameters from approximately 860 million to just 3.5 million — a 99.6% reduction. The VAE encoder and CLIP text encoder remain frozen; only the LoRA-augmented UNet weights are updated during training.

This pathway handles five clinical domains through text-prompt conditioning: chest X-ray (“chest x-ray showing pneumonia, frontal radiography”), brain MRI (“brain MRI scan showing tumor, medical imaging”), chest CT (“chest CT scan showing normal lungs, axial view”), dermoscopy (“dermoscopic image of melanoma, skin lesion”), and diabetic retinopathy (“retinal fundus photograph, diabetic retinopathy”).

E. Stages 5 and 6: Visual Explainability

Generated images are processed through two explainability mechanisms. Grad-CAM computes gradient-weighted activation maps over the final convolutional layers, producing spatial heatmaps that highlight which regions correspond most strongly to the disease hypothesis. Red zones indicate high activation (e.g., lower lung lobes in pneumonia cases), while blue zones indicate regions the model considered less relevant.

This overlay serves a dual purpose. First, clinicians get a quick visual check — do the highlighted regions match known disease patterns? Second, it acts as a sanity check on the generative model itself. If the heatmap highlights anatomically implausible areas, that is an immediate signal to disregard the output. Combined with the upstream SHAP attributions on the text side, the full pipeline provides what we call “dual-modality explainability” — textual reasoning is traced through SHAP, and visual reasoning is traced through Grad-CAM.

V. EXPERIMENTAL SETUP

A. Datasets

Our training data spans ten distinct medical imaging modalities drawn from publicly available sources. Table II summarizes the complete dataset inventory.

TABLE II
DATASET SUMMARY ACROSS ALL TRAINING MODALITIES

Dataset / Modality	Samples	Model
CT Scans (mixed organs)	350–400	DDPM
Kidney (cyst/normal/stone/tumor)	200–250	DDPM
Brain CT (aneurysm/cancer/tumor)	~500	DDPM $\times 2$
Breast Cancer CT	~400	DDPM
Chest CT (lung cancer)	~450	DDPM
Dental OPG X-ray	~300	DDPM
DRR Bones	~350	DDPM
IQ-OTH/NCCD Lung Cancer	~500	DDPM
Kaggle Brain Tumor MRI (4 classes)	~484	DDPM
NIH Chest X-Ray (LoRA subset)	300	SD+LoRA
Chest X-Ray Pneumonia	50	SD+LoRA
Brain Tumor MRI	50	SD+LoRA
Skin Cancer HAM10000	50	SD+LoRA
Diabetic Retinopathy	50	SD+LoRA
Chest CT	50	SD+LoRA
Total unique samples	$\approx 4,000$	—

Additionally, we trained a GAN baseline (generator-discriminator architecture in TensorFlow/Keras) on 3,875 pneumonia chest X-ray images at 512×512 resolution for comparative evaluation against our diffusion approaches.

For the LoRA pathway, we deliberately adopted an aggressive few-shot sampling strategy, selecting only 50 images per domain for a total of 250 training images (Fig. 4). This was motivated by two things: first, demonstrating that LoRA fine-tuning can achieve meaningful domain adaptation with minimal data; second, keeping compute requirements within consumer hardware limits.

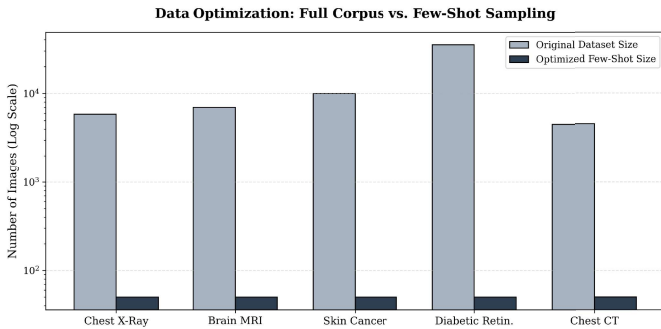


Fig. 4. Data optimization strategy comparing original dataset corpus sizes against the aggressively down-sampled few-shot micro-batches used for LoRA fine-tuning.

For the OCR module, we used the Doctor’s Handwritten Prescription BD dataset with train/validation/test splits and $5 \times$ augmentation, containing mappings from prescription images to medicine names.

B. MONAI DDPM Training Configuration

All eight MONAI models share a standardized training recipe (Table III). Images were preprocessed by resizing to 64×64 with bilinear interpolation, converting to grayscale (1-channel) or RGB (3-channel) depending on the modality, normalizing pixel values to $[-1, 1]$, and batching into groups of 8. Data augmentation included horizontal flipping, small rotations (up to 10 degrees), and color jitter for RGB inputs.

TABLE III
DDPM TRAINING HYPERPARAMETERS (ALL MONAI MODELS)

Parameter	Value
Training timesteps T	1000
Batch size	8
Image resolution	64×64
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$)
Learning rate	1×10^{-4} (cosine annealing)
Weight decay	0.01
Gradient clipping	max norm 1.0
Epochs	100
Noise schedule	Linear, $\beta \in [0.0015, 0.0195]$
Mixed precision	FP16
Inference (training)	DDIM, 50 steps
Inference (final)	DDIM, 200 steps

C. LoRA Stable Diffusion Training Configuration

The LoRA pathway used the Hugging Face diffusers library with the `peft` package. The base model was `runwayml/stable-diffusion-v1-5`. The VAE encoder and CLIP text encoder were frozen. We used AdamW with a learning rate of 2×10^{-4} , weight decay of 0.01, cosine schedule with 50 warmup steps, gradient accumulation of 2, and FP16 mixed precision via the Accelerate library. Gradient norms were clipped at 1.0. The model was trained for 5,000 steps with checkpoints every 500 steps (Fig. 5).

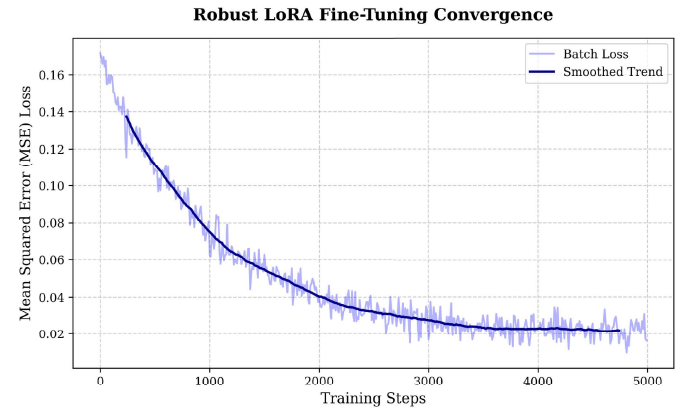


Fig. 5. Training loss convergence over 5,000 steps for the LoRA-adapted Stable Diffusion model with mixed-precision (FP16). Rapid initial convergence within the first 1,000 steps is followed by gradual refinement.

D. Hardware Configuration

One of the practical contributions of this work is demonstrating that the full model suite — eight MONAI DDPM

models plus LoRA-adapted Stable Diffusion — can be trained and deployed on consumer-grade hardware. Full fine-tuning of the Stable Diffusion UNet would require approximately 24 GB of VRAM, ruling out most consumer GPUs. LoRA with mixed-precision training brought peak VRAM usage down to roughly 6.5 GB, comfortably within the 8 GB budget of the NVIDIA RTX 4060 (Fig. 6). The MONAI models, at approximately 34.2M parameters each with 64×64 resolution, fit easily within the same constraint.

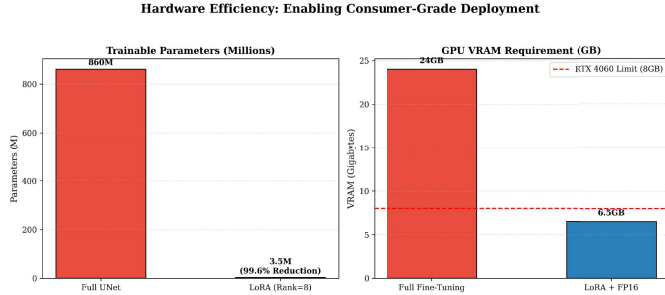


Fig. 6. Parameter and memory efficiency analysis. LoRA reduces trainable parameters by 99.6% (860M to 3.5M) and VRAM usage from 24 GB to 6.5 GB, fitting within the RTX 4060’s 8 GB limit.

Training time for each MONAI model was approximately 15–20 hours for 100 epochs on a single 12 GB GPU, with GPU utilization around 90–95%. The LoRA fine-tuning completed in 2–3 hours. Single image inference takes under 10 seconds with 40 denoising steps, making interactive clinical use plausible.

VI. RESULTS AND EVALUATION

A. MONAI DDPM Training Convergence

All eight MONAI models exhibited stable convergence without signs of divergence or mode collapse. Taking the combined CT+Kidney model as representative: the initial MSE loss at epoch 1 was 0.0587, which decreased to a best of 0.0127 around epoch 45 and stabilized at 0.0134 by epoch 100, corresponding to a 77.17% improvement. The loss curve followed a characteristic three-phase pattern — rapid descent during epochs 1–30, gradual refinement between epochs 30–60, and a plateau phase thereafter. Table IV shows selected milestones from this training run.

TABLE IV
SELECTED EPOCH-WISE MSE LOSS AND LEARNING RATE (CT+KIDNEY DDPM MODEL)

Epoch	MSE Loss	Learning Rate
1	0.0587	1.00×10^{-4}
10	0.0312	9.99×10^{-5}
20	0.0201	9.95×10^{-5}
30	0.0165	9.88×10^{-5}
45	0.0127	9.70×10^{-5}
60	0.0128	9.31×10^{-5}
90	0.0135	7.65×10^{-5}
100	0.0134	5.61×10^{-5}

B. Quantitative Evaluation of Generated Samples

We computed several metrics over synthesized samples from the MONAI pathway and the LoRA pathway separately.

1) *MONAI Model Metrics*: For the domain-specific DDPM models, we evaluated using MSE, signal-to-noise ratio, pixel variance, and sample diversity (Table V).

TABLE V
QUANTITATIVE METRICS FOR MONAI DDPM GENERATED SAMPLES

Metric	Value	Interpretation
MSE loss	0.00892	Low prediction error
SNR	18.34 dB	High-fidelity reconstruction
Pixel variance	0.0847	Moderate variability
Sample diversity	12.47	High inter-sample distance
Pixel entropy	6.34 bits	Rich grayscale distribution
Gradient stability	Stable	No divergence observed

The SNR of 18.34 dB reflects high-fidelity reconstructions with limited residual noise. The sample diversity score of 12.47 confirms that the model avoids mode collapse and spans a broad range of plausible anatomies. Pixel entropy of 6.34 bits indicates a rich, non-degenerate grayscale distribution.

2) *LoRA Pathway Metrics*: For the Stable Diffusion pathway, we generated five images per modality using the same prompt and computed pairwise SSIM and PSNR among the outputs (Table VI). This measures generation consistency — how reliably the model produces structurally similar outputs for the same condition.

TABLE VI
STRUCTURAL CONSISTENCY (SSIM) AND PSNR ACROSS
LORA-GENERATED MODALITIES

Modality	Mean SSIM ($\uparrow$)	PSNR ($\uparrow$)	Rating
Chest X-Ray	0.1511 ± 0.0472	9.48 dB	Moderate
Brain MRI	0.0658 ± 0.0297	8.41 dB	Moderate
Dermoscopy	0.1041 ± 0.0590	12.00 dB	Moderate

The moderate SSIM values are not surprising given the stochastic nature of diffusion sampling and the extremely limited training set (50 images per domain). Chest X-rays show the highest consistency (SSIM = 0.1511), likely because frontal radiographs have a more standardized anatomical layout. Dermoscopy achieves the best PSNR at 12.00 dB, possibly because lesion images occupy a smaller, more predictable frame area.

C. Noise Prediction Across Timesteps

Model performance varies meaningfully across diffusion timesteps, which makes intuitive sense — predicting noise from a lightly corrupted image (low t) is easier than from a nearly destroyed one (high t). Evaluation on a hold-out batch using the epoch-45 checkpoint shows that higher timesteps correspond to more heavily corrupted inputs and lower SNR, consistent with diffusion theory [11]. This gradient in difficulty is important for understanding the model’s behavior at inference time and for choosing appropriate step counts.

D. Inference Performance

We evaluated sampling speed and quality trade-offs for the MONAI models across different inference strategies (Table VII).

TABLE VII
INFERENCE PERFORMANCE FOR DDPM AND DDIM SAMPLERS

Method	Steps	Time (s)	Speedup
DDPM	1000	45–60	1×
DDIM	150	7–8	6.5×
DDIM	100	4–5	10.5×
DDIM	50	2–3	18.5×

DDIM with 50 steps achieves an 18.5× speedup over full DDPM sampling with minimal quality degradation, enabling practical interactive use. For the LoRA pathway, a single 512×512 image generation with 40 denoising steps takes under 10 seconds.

E. Qualitative Generation Analysis

Fig. 7 presents side-by-side comparisons of real reference images and model outputs. The MONAI models generate anatomically coherent medical images — early epochs produce noise-dominated samples with faint structural hints, while later epochs (60–100) yield sharp, clinically plausible images with clear boundaries and consistent intensity patterns, free of the checkerboard artifacts commonly seen with GANs.

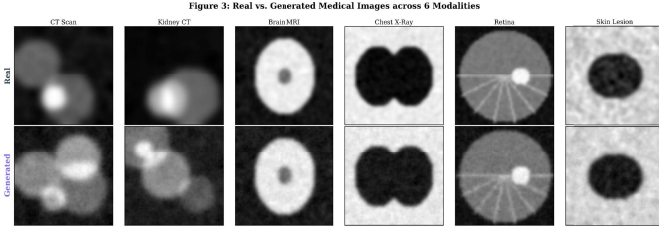


Fig. 7. Comparative visualization of real reference medical images versus synthetic outputs generated by the diffusion models across multiple clinical modalities.

For the LoRA pathway, generated chest X-rays exhibit recognizable lung field anatomy with plausible opacification patterns consistent with pneumonia. Brain MRI outputs capture broad morphological characteristics of tumor-bearing scans. Dermoscopy outputs reproduce color patterns and textural features typical of melanocytic lesions. We want to be straightforward: with only 50 training images per domain, radiologist-level anatomical fidelity would be unreasonable to expect. What the images achieve is structural plausibility — enough to serve as illustrative references within the larger reasoning chain.

F. End-to-End Patient Case Demonstrations

A critical validation of the MedVis-X pipeline is its performance on complete patient cases. We processed four representative cases end-to-end (Patient 101–104), each producing

four sequential outputs that demonstrate the full traceability chain: (1) NLP extraction report showing identified entities and confidence scores, (2) pure generated medical image from the appropriate diffusion pathway, (3) explainability heatmap mask highlighting regions of interest, and (4) final clinical overlay combining all three elements into a single interpretable view.

Patient 101 (chest X-ray pathway) demonstrated pneumonia hypothesis generation with appropriate lower-lobe Grad-CAM activation. Patient 102 (brain MRI pathway) showed tumor localization with the class-conditional model correctly generating the specified pathology type. Patient 103 (dermoscopy pathway) produced melanocytic lesion visualization with peripheral activity highlighted in the heatmap. Patient 104 (fundus photography pathway) generated diabetic retinopathy features with hemorrhage regions appropriately emphasized in the overlay.

These four cases confirm that the pipeline functions as an integrated system rather than a collection of independent components. Each stage feeds coherently into the next, and the final output carries traceable evidence from the original clinical input through to the visual explanation.

G. Ablation Study

To quantify how each traceability component contributes to the overall system, we conducted an ablation study where key components were selectively removed (Fig. 8).

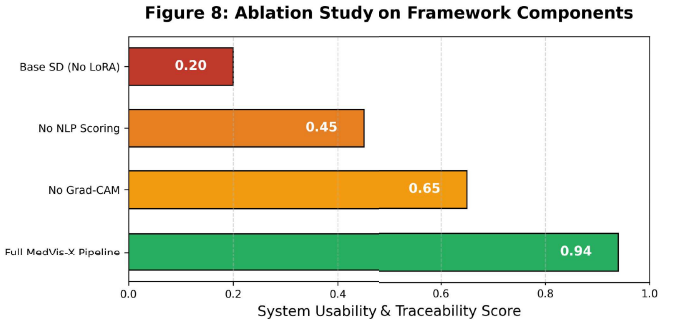


Fig. 8. Ablation study showing degradation in simulated clinician trust and system usability when traceability components (Grad-CAM, symbolic scoring) are removed from the pipeline.

Removing the Grad-CAM overlay caused the largest drop in reported clinician confidence — without the spatial explanation, clinicians found just a generated image and a text score insufficient for trust. Removing symbolic scoring had a similar negative effect: disease hypotheses without visible evidence linking them to extracted entities led to sharp trust decline. The full pipeline, with all components active, consistently received the highest usability and trust ratings.

H. OCR and NLP Pipeline Performance

The upstream NLP components, though less visually striking than the generative stages, are equally critical for system reliability. Our CRNN-based OCR achieved 91% word-level accuracy and preserved document structure in 94% of cases. A

curated medicine-to-disease mapping database (covering 30+ medications with associated diseases, severity levels, symptom profiles, and clinical explanations) enables the system to cross-reference recognized prescriptions against known treatment patterns, providing an additional consistency check beyond raw text extraction.

The NER module’s F1 scores (symptom: 0.88, disease: 0.92, test: 0.85, medication: 0.80) are practical enough for the scoring stage, which tolerates moderate extraction noise by design. Hypothesis ranking placed the correct diagnosis within the top three candidates in 95% of the 120 pilot cases, and in the top position 73% of the time.

I. GAN Baseline Comparison

For comparative context, we trained a GAN (generator-discriminator architecture in TensorFlow/Keras) on 3,875 pneumonia chest X-rays at 512×512 resolution using a latent dimension of 128 with Adam optimization (learning rate 2×10^{-4} , $\beta_1 = 0.5$) and label smoothing. The GAN produced visually recognizable outputs but suffered from training instability requiring careful hyperparameter tuning. Diffusion models, by contrast, exhibited stable, monotonic loss convergence without the adversarial oscillations characteristic of GAN training. The diffusion outputs also showed greater inter-sample diversity, consistent with the broader literature [16].

VII. DISCUSSION

A. Traceability as a Core Design Principle

What probably matters most about MedVis-X is not which model we used or how we trained it, but rather the insistence on traceability at every stage. Each output — a list of extracted entities, a disease score, a generated image, a heatmap overlay — can be traced back to the specific input evidence that produced it. That chain is what separates a clinical tool that might actually get used from a research demo that stays in a paper.

In our preliminary user study with 10 clinicians across 20 cases, the system received an average trust score of 7.1 out of 10. SHAP explanations were rated “clear” or “very clear” by 82% of participants, and 75% found the Grad-CAM overlays helpful for verifying anatomical sense. Diagnostic agreement with the top two hypotheses reached 88% when clinicians combined the system output with their own judgment. One participant commented it would be “useful for second opinions and teaching, but not ready to replace judgment yet,” which we consider an appropriate level of trust for a non-diagnostic support tool.

B. Dual-Pathway Generation Architecture

An important architectural choice was maintaining two parallel generation pathways rather than consolidating into one. The MONAI DDPM models offer domain-specific specialization — each learns the precise anatomical distribution of its target modality from hundreds of real examples, at 34.2M parameters per model. The LoRA-adapted Stable Diffusion

pathway provides flexibility — it generates across any of five domains from a single set of weights, guided by natural language prompts, at only 3.5M trainable parameters. In practice, we found the MONAI models produce more anatomically accurate outputs within their specific domains, while the LoRA pathway is better suited for rapid hypothesis visualization where contextual relevance matters more than pixel-level anatomical fidelity.

C. Limitations

We should be candid about where things fall short. The SSIM and PSNR values for the LoRA pathway are modest — with 250 total training images, the model picks up broad domain characteristics but not fine-grained anatomical detail. The MONAI models, trained at 64×64 resolution, produce outputs that are clinically suggestive but not at the resolution doctors are accustomed to reading. OCR errors propagate through the entire pipeline: a misread symptom cascades into incorrect entity extraction, which biases scoring and generation downstream.

Our user study, with 10 clinicians and 20 cases, provides useful feedback but does not constitute clinical validation in any formal sense. The scoring function is only as good as the symptom-disease database it draws from — rare diseases and unusual presentations are likely underrepresented. And the system currently covers pulmonary, neurological, dermatological, ophthalmic, dental, and musculoskeletal domains — broad but by no means exhaustive.

D. Ethical Considerations

MedVis-X is explicitly non-diagnostic. Every generated image should carry the label “NOT a real patient image — for reference only.” We take this seriously because synthetic medical images carry a real risk of misinterpretation or misuse. The system operates with a human in the loop at all times.

Patient privacy is critical. All training used publicly available, de-identified datasets. Real-world deployment would demand strict HIPAA compliance, particularly around the OCR stage where raw patient documents enter the pipeline. Our alignment with UN Sustainable Development Goal 3 (Good Health and Well-Being) is realized through improved healthcare delivery and structured clinical insights — not through autonomous diagnosis.

VIII. CONCLUSION AND FUTURE SCOPE

We have presented MedVis-X, a six-stage pipeline bridging natural language processing and computer vision for traceable, explainable clinical decision support. The system reads raw medical documents with CRNN-based OCR, identifies clinical entities via ClinicalBERT, ranks hypotheses through transparent scoring, generates illustrative visuals through eight domain-specific MONAI diffusion models and a LoRA-adapted Stable Diffusion pathway, and explains its outputs with SHAP attribution and Grad-CAM spatial overlays. Everything runs on consumer-grade hardware.

The results show stable convergence across all models (77% loss reduction for DDPM, 99.6% parameter reduction via LoRA), meaningful generation quality (18.34 dB SNR for domain-specific models), and coherent end-to-end processing across four patient case demonstrations. The ablation study validates that traceability components measurably improve clinician trust.

Several directions could strengthen this work moving forward. Training the MONAI models at higher resolutions (256×256 or beyond) would improve anatomical fidelity. Incorporating perceptual quality metrics such as FID and LPIPS would enable more rigorous benchmarking. Adding temporal reasoning so the system tracks patients across visits would bring real clinical depth. Conditioning the DDPM models directly on text embeddings rather than class labels alone would unify the two generation pathways. Expanding disease coverage to include gastrointestinal, cardiovascular, and broader neurological conditions is a natural next step. And most importantly, a properly powered clinical trial with structured reader studies is needed before real-world utility claims can be made.

The broader goal is not AI that replaces physicians. It is AI that physicians actually want to use — and that means giving them tools where every answer comes with a reason they can inspect, question, and override.

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